

Polite on the Surface, Broken in Practice: A Curated Dataset for Fixing Generation and Register Failures in Low-Resource Bangla Text Generation

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Abstract

Recent advances in Multilingual Large Language Models (MLLMs) have improved cross-lingual conversational capabilities, yet generating reliable, institutionally usable text in low-resource languages like Bangla remains a critical bottleneck, particularly for compact, low-resource LLMs. Beyond surface fluency, state-of-the-art and small open-weight models frequently exhibit deeper generation failures in Bangla: incoherent or off-topic output, structural misformatting, and inconsistent honorific registers, that render text unusable in formal contexts. To address this, we introduce **BLADE (BangLa Application and DialoguE generation)**, a culturally aligned instruction-tuning dataset and benchmarking framework comprising 4,196 curated interaction pairs. We use this resource to fine-tune and evaluate compact open-weight architectures, including DeepSeek-8B, LLaMA-3.2-3B, Qwen2, and TigerLLM-1B, via LoRA adapters under 4-bit NormalFloat (NF4) quantization on limited hardware. Our evaluations show that targeted fine-tuning on BLADE yields substantial improvements in generation reliability, structural fidelity, and register consistency even for the smallest models, establishing a rigorous benchmark for closing generation and pragmatic gaps in low-resource multilingual text generation. Code and dataset: https://github.com/ashuvo25/Bangla_Application_LLM/tree/main

1 Introduction

Large Language Models (LLMs) have fundamentally transformed multilingual text generation (Achiam et al., 2023; Touvron et al., 2023), yet for low-resource languages this progress remains superficial, and the gap widens further for *low-resource LLMs*, compact, open-weight models adapted under limited compute and data. The multilingual curse manifests not merely as vocabulary gaps, but as a deeper generation reliability gap: the

Zero-Shot (Flawed)

...প্রাইডা, সাধারণত ১৫-
১৭.০২.২০২৩ | শুখ:
উচ্চকেন্দ্র বিবর্তন: ঢঙের
ঘনমঞ্জর... ভারসাম্য:
-২০□সে/-৪□ফা...

(English: fragmented, off-topic output unrelated to the "leave application" prompt.)

BLADE-SFT (Correct)

তারিখ: ১৮/১০/২০২৪
প্রিঃ ... বিষয়ঃ অগ্রিম
ছুটির জন্য প্রধান শিক্ষকের
নিকট আবেদন। মহোদয়,
সম্মানপূর্বক বিনীত নিবেদন
এই যে...

(English: coherent, on-topic, correctly structured formal application.)

Table 1: Example of generation incoherence in a compact zero-shot LLM (LLaMA-3.2-3B) versus the corrected output after BLADE fine-tuning, for the prompt "Application for advance leave."

divide between surface fluency and functional usability. This gap surfaces as outright generation failures, incoherent, off-topic, or structurally broken output, compounded by subtler pragmatic failures such as inconsistent honorific registers, both of which render text institutionally unusable.

Bangla, the fifth most spoken language worldwide with over 230 million native speakers, makes this failure concrete. Current multilingual models, and especially compact low-resource LLMs commonly deployed for low-resource-language tasks, produce Bangla text ranging from incoherent or off-topic to merely superficially fluent: misplaced document structures, inconsistent honorific registers, degenerate content, and culturally incongruent discourse markers that make outputs institutionally unacceptable (Mukherjee et al., 2025; Snigdha and Rahman, 2022).

The root causes are twofold. First, compact low-resource LLMs, often the only computationally feasible option for low-resource-language deployment, are prone to producing incoherent, off-topic, or degenerate output on structured, format-sensitive tasks without targeted supervision. Sec-

ond, even fluent output frequently lacks *register-aware supervision*: Bangla encodes social relationships directly into its grammar through a two-tier honorific system, the formal আপনি (Apni: You) obligatory in institutional writing, versus the informal তুমি (Tumi: You) reserved for peers and juniors 8. Mixing these registers signals a fundamental breakdown in social competence and renders a document unusable in any formal context (Snigdha and Rahman, 2022). Table 1 illustrates the former: a compact zero-shot model produces text entirely unrelated to the prompt rather than a merely mis-registered response.

To address these limitations, we introduce BLADE (BangLa Applications and DialoguEs), a specialized resource serving a dual role: (i) an instruction-tuning dataset of 4,196 high-quality, expert-annotated prompt-response pairs spanning 2,008 unique topics, and (ii) a rigorous evaluative benchmark for measuring model adherence to Bangla generation reliability, structural, and cultural norms across both large multilingual models and compact low-resource LLMs. BLADE was constructed through a three-tier acquisition pipeline: government-approved textbooks establish canonical formatting; verified web portals provide real-world usage diversity; and author-synthesized examples, cross-validated by native linguistic experts, address complex generation and honorific alignment scenarios underrepresented in static sources.

Our contributions are as follows: (1) We introduce BLADE, a publicly available instruction-tuning dataset of 4,196 expert-annotated pairs spanning 2,008 unique topics, built through a three-tier pipeline with native linguistic expert validation, covering educational, professional, and conversational Bangla generation. (2) We benchmark generation reliability and pragmatic gaps in five state-of-the-art multilingual LLMs and four compact low-resource LLMs under zero-shot conditions, systematically documenting coherence failures, register inconsistencies, and structure violations across all tested architectures. (3) We demonstrate that targeted SFT on BLADE yields consistent gains across automatic metrics (BLEU 17.73; chrF 46+), human expert evaluation (>4.6/5 on all dimensions), and LLM-as-judge scoring (8.30/10), establishing that cultural data specificity outweighs model scale for reliable, low-resource pragmatic generation.

2 Related Work

Multilingual LLMs and the Low-Resource Gap.

Progress in LLMs has been driven by web-scale corpora and instruction datasets such as Common Crawl, Alpaca, and Dolly, with multilingual coverage expanded by projects like HPLT (de Gibert et al., 2024). Despite these gains, a well-documented 25–35% instruction-following accuracy gap persists between high- and low-resource languages (Zeng et al., 2025), and LLMs consistently fail to outperform classical baselines on extremely low-resource languages in zero-shot settings (Cahyawijaya et al., 2024). Critically, machine-translated instruction data systematically underestimates model capability compared to natively localized equivalents (Bawden and Yvon, 2023), motivating our choice to construct BLADE from human-curated, institutionally grounded Bangla sources.

Honorific Systems and Pragmatic Competence.

A deeper multilingual failure concerns pragmatic competence: generating text that is socially and institutionally appropriate, not merely grammatical. For languages that grammaticalize social relationships into morphology, register consistency is a necessary condition for functional usability. Japanese LLMs fail to generalize honorific patterns across novel syntactic structures (Sekizawa and Yanaka, 2023), and even frontier models including GPT-4o underperform on high-politeness Javanese registers underrepresented in web-scale data (Farhansyah et al., 2025). Arabic evaluation reveals the same failure: surface fluency does not imply cultural coherence (Attia et al., 2026).

These findings establish that Bangla’s honorific mismatch problem, discussed above, is a cross-linguistically shared structural failure, not a Bangla-specific anomaly.

Bangla NLP Resources. Bangla NLP has progressed from encoder models such as *BanglaBERT* (Bhattacharjee et al., 2022) to native generative models including *TigerLLM* (Raihan and Zampieri, 2025) and *TituLLMs* (Nahin et al., 2025), alongside task-specific datasets for NER (Paul et al., 2026; Mhaske et al., 2023), captioning (Rahman et al., 2019), and sentiment analysis (Islam et al., 2021). However, all existing resources target classification and extraction; none provides supervision for long-form, format-sensitive, register-consistent generation the precise gap BLADE addresses.

Topic	Content (Bangla)	English Translation
অসুস্থতার কারণে ছুটির আবেদন (Application for sick leave)	তারিখঃ ১৮/১০/২০২৪ খ্রিঃ বরাবর প্রধান শিক্ষক হোলমাইদ হাই স্কুল এন্ড কলেজ ভাটারা, ঢাকা-১২১২ বিষয়ঃ বিদ্যালয়ে অনুপস্থিতির জন্য প্রধান শিক্ষকের নিকট আবেদন। স্যার	Date: 18/10/2024 AD To Headmaster Solmaid High School and College Vatara, Dhaka-1212 Subject: Application to the Headmaster re- garding absence from school. Sir:.....

Table 2: Example BLADE entry with English translation, illustrating canonical document structure.

Data Quality over Scale. Our core empirical finding that a compact model fine-tuned on BLADE outperforms zero-shot frontier models is grounded in a broader principle: Zhou et al. (2023) show that 1,000 carefully curated examples suffice to match models trained with extensive RLHF, establishing that data quality outweighs quantity for alignment. This has been validated in low-resource multilingual settings, where small-scale high-quality SFT data outperforms larger automatically constructed corpora at the morphological and structural level (Iyer et al., 2024). BLADE provides concrete evidence of this principle for register-sensitive, format-constrained Bangla generation.

3 BLADE Dataset

3.1 Dataset Overview

The BLADE dataset comprises 4,196 high-quality prompt-response pairs constructed specifically for structured Bangla generation, with an average response length exceeding 1,300 tokens. The dataset spans 2,008 unique topics across two primary task types application writing and dialogue generation covering educational, professional, administrative, and conversational domains Table 3. BLADE was designed around three core principles absent from existing Bangla corpora: (i) register awareness every example enforces consistent honorific alignment between salutations, pronouns, verbal morphology, and closings; (ii) structural fidelity application examples strictly follow canonical Bangla institutional document order (Date → Addressee → Subject → Salutation → Body → Closing → Signature); and (iii) topical diversity topics represent real-world Bangla institutional writing encountered by students, professionals, and citizens in Bangladesh. An illustrative entry is provided in Table 2.

Applications		Dialogues	
Topic	Count	Topic	Count
General	675	General	476
Educational	214	Informal	170
Administrative	169	Formal	141
Professional	163		
Total Topics			2,008

Table 3: Distribution of unique topics across domains.

3.2 Topic Selection

Topic inclusion followed four criteria: (i) *real-world prevalence* in Bangladeshi educational and professional contexts, validated against source frequency across textbooks and web portals; (ii) *register sensitivity* tasks where honorific errors render output institutionally unusable; (iii) *structural complexity* multi-component documents beyond a single paragraph; and (iv) *under representation* in existing pretraining corpora. Topics already well-covered by generic web-scale data (e.g., news, Wikipedia) were explicitly deprioritized. The resulting distribution spans 1,221 application topics and 787 dialogue topics Table 3.

3.3 Data Collection

Collection followed a three-tier acquisition strategy, each tier governed by distinct sourcing and verification protocols. Full annotation guidelines, annotator profiles, inter-annotator agreement statistics, and quality control criteria are provided in Appendix A.

Tier 1: Institutional Textbooks (1,972 samples).

Canonical application structures were extracted from nine government-approved NCTB secondary and higher secondary textbooks, establishing a gold standard for institutional formatting. Each en-

try was verified by two independent annotators for structural completeness (all seven mandatory document components present in canonical order) before inclusion.

Tier 2: Public Web Portals (1,382 samples).

Real-world examples were curated from 14 verified Bangla educational web portals Table 10. Every entry underwent manual verification against structural completeness and honorific consistency criteria. Approximately 23% of candidate web entries were discarded during this process due to register inconsistencies or malformed structures.

Tier 3: Author-Synthesized Examples (842 samples).

To address complex scenarios underrepresented in static sources particularly multi-turn dialogues requiring sustained honorific consistency 842 examples were purpose-built by the authors and cross-validated by two external native linguistic experts specializing in Bangla philology. Each Tier 3 example passed a two-stage review: author-annotator drafting followed by independent expert adjudication.

3.4 Data Preprocessing

Raw entries are cleaned using regular expressions preserving Bangla Unicode (U+0980–U+09FF), reformatted into the instruction-tuning template described in Section 4.2, and partitioned into a 80% training set 3,356 examples 10% testing set 420 examples and 10% validation set 420 examples via stratified sampling across domain categories with a fixed random seed. The complete pipeline is illustrated in Figure 1.

4 Methodology

Here we describe the experimental setups for both zero-shot evaluation and supervised fine-tuning.

4.1 Zero-Shot Setup

To establish a baseline for multilingual LLMs, we conducted a zero-shot evaluation on our dataset. The dataset coverage all domain categories Table 3. Each model was queried using the identical structured template described in Section 4.2 to isolate true instruction-following capability without task-specific demonstrations. It is important to note that zero-shot models are evaluated on a prompt format they have not been explicitly trained on; consequently, some portion of the observed performance gap may reflect *format unfamiliarity* rather than Bangla specific incapability alone.

For inference, we utilized the Groq API¹ to evaluate Llama-4-Scout-17B, Gemma2-9B, and Kimi-K2-32B. Google AI Studio was used to evaluate Gemini-2.5-Flash and Gemini-2.0-Flash. Performance was benchmarked against BLADE ground-truth responses.

4.2 Fine-tuning Setup

To adapt pre-trained LLMs for Bangla instruction following, we employed Parameter-Efficient Fine-Tuning (PEFT) using 4-bit NormalFloat (NF4) quantization and Low-Rank Adaptation (LoRA) (Hu et al., 2021). Use five models DeepSeek-R1-Distill-Llama-8B, Qwen2-(0.5–1.5)B, Llama-3.2-3B-Instruct, and TigerLLM-1B-it to evaluate BLADE across varying scales and training paradigms. TigerLLM serves as a Bangla-native baseline; the remaining models represent state-of-the-art multilingual architectures at the 1B–8B scale, chosen to test efficacy under resource-constrained deployment conditions.

All models were loaded in 4-bit precision with FP16 computation via the Unsloth library on dual NVIDIA Tesla T4 GPUs (32 GB total VRAM). SFT was conducted for 2 epochs with a 2048 token sequence length using SFTTrainer, batch size 2, cosine learning rate decay with 5 warm-up steps at peak 2×10^{-5} , and AdamW optimizer. LoRA was configured with rank $r=16$, $\alpha=32$, dropout 0.05, applied to both attention and MLP projection layers. The prompt template have (Listing 4.2) enforced canonical document order (Subject → Salutation → Body → Closing) and register consistency, with loss computed only over the `<assistant>` span.

Prompt Template

```
<system>
order: Subject → Salutation → Body →
Closing. Keep consistent honorifics.
<user>
Task: Formal application.
Subject: অগ্রিম ছুটির জন্য আবেদন
Addressee: প্রধান শিক্ষক, এক্সওয়াইজেড উচ্চ বিদ্যালয়
Applicant: মোঃ রহমান, নবম শ্রেণি, রোল ১২
Reason: ডাক্তার দেখাতে শহরে যেতে হবে
Dates: ১০-১২ আষাঢ় ১৪৩২
Closing: বিনীতভাবে
Constraints: 150--200 words;Bangla
digits;nocode-mixing.
<assistant>
```

¹<https://console.groq.com/>

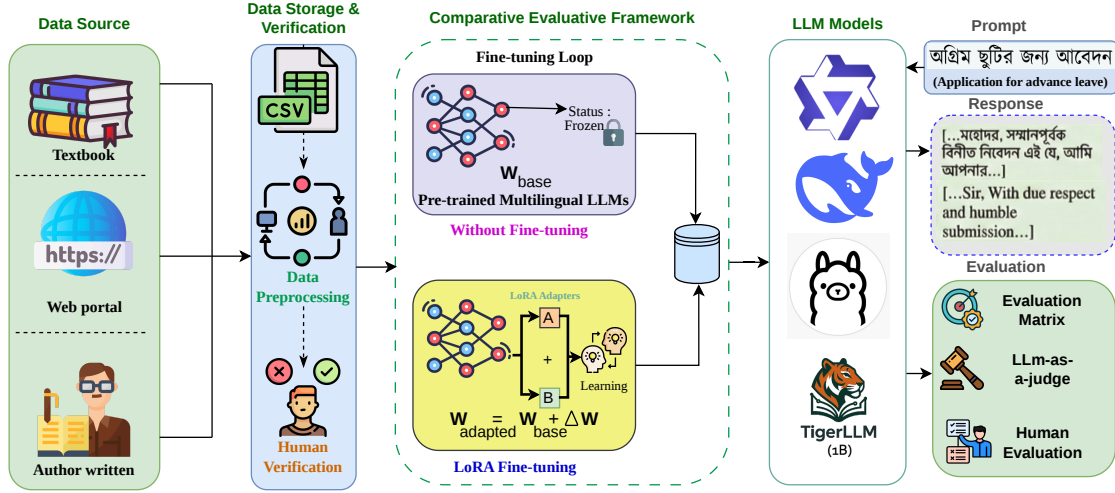


Figure 1: BLADE methodology: three-tier data collection, LoRA-based fine-tuning of pre-trained multilingual LLMs, and a three-way comparative evaluation framework

4.3 Evaluation Metrics

We evaluate model outputs using five complementary metrics that collectively capture lexical, structural, and semantic similarity between generated and reference texts. BLEU (Papineni et al., 2002) measures n -gram overlap with a brevity penalty. chrF (Popović, 2015) computes a character n -gram F-score, making it particularly robust for morphologically rich languages like Bangla. ROUGE-L (Lin, 2004) measures the longest common subsequence between reference and hypothesis. WER (word error rate) measures normalized edit distance between hypothesis and reference, where lower scores indicate fewer structural errors. BERTScore (Zhang et al., 2019) evaluates semantic similarity using contextual embeddings from a pretrained model. Given the known limitations of similarity-based metrics in capturing semantic coherence and cultural alignment, automatic metric results are complemented by **human expert** evaluation and **LLM-as-judge** scoring in Section 5.

4.4 LLM-as-Judge Procedure

To assess functional usability beyond automatic metrics, we employed GPT-4.1 as an LLM judge across 420 randomized samples. The judge was provided with the input prompt (Appendix B), the rationale for selecting GPT-4.1 (Appendix B.2), a reference response, and the model output, and instructed to score on a 1–10 scale against three criteria: (i) *structural correctness* presence and ordering of all mandatory document components; (ii)

register consistency honorific alignment throughout; and (iii) *semantic relevance* whether the output addresses the stated prompt purpose. The judge prompt was fixed across all 420 samples to ensure scoring consistency, and outputs were parsed programmatically; malformed scores were re-queried once before exclusion.

4.5 Human Evaluation

Three native Bangla-speaking philology experts conducted a double-blind assessment of 100 stratified samples, rating outputs on a 5-point Likert scale across: (i) *Structural Integrity* adherence to canonical document formatting; (ii) *Fluency* grammatical correctness and naturalness; and (iii) *Cultural Alignment* honorific register consistency. Inter-annotator agreement was robust (Spearman’s $\rho=0.84$, Kendall’s $\tau=0.76$), and post-SFT scores exceeded 4.6/5 across all dimensions (Table 5), confirming that BLADE fine-tuning enables models to produce institutionally appropriate, functionally usable Bangla documents.

5 Result Analysis

We benchmark our dataset through two assessments: (1) evaluating state-of-the-art models in a zero-shot setting and (2) comparing model performance before (base model) and after Supervised Fine-Tuning (SFT). This dual approach highlights the dataset’s utility and the knowledge gap it addresses in NLP.

Model	Condition	BLEU (↑)	ROUGE-L (↑)	chrF (↑)	WER (↓)	BERTScore (↑)
Kimi-K2-32B	Zero-Shot	4.55	0.670	37.15	2.283	0.722
Kimi-K2-32B	FM	5.12 _{↑12.5%}	0.731 _{↑9.1%}	40.38 _{↑8.7%}	1.974 _{↓13.5%}	0.739 _{↑2.4%}
LLaMA-4-Scout-17B	Zero-Shot	6.30	0.290	35.92	1.729	0.715
LLaMA-4-Scout-17B	FM	7.43 _{↑17.9%}	0.348 _{↑20.0%}	39.11 _{↑8.9%}	1.521 _{↓12.0%}	0.728 _{↑1.8%}
Gemini-2.0-Flash	Zero-Shot	3.73	0.250	34.45	3.895	0.711
Gemini-2.0-Flash	FM	4.58 _{↑22.8%}	0.317 _{↑26.8%}	38.27 _{↑11.1%}	3.114 _{↓20.1%}	0.726 _{↑2.1%}
Gemma2-9B	Zero-Shot	4.69	0.100	34.80	1.765	0.706
Gemma2-9B	FM	5.47 _{↑16.6%}	0.134 _{↑34.0%}	37.93 _{↑9.0%}	1.483 _{↓16.0%}	0.719 _{↑1.8%}
Gemini-2.5-Flash	Zero-Shot	7.88	0.250	41.04	1.934	0.728
Gemini-2.5-Flash	FM	9.14 _{↑16.0%}	0.390 _{↑56.0%}	45.67 _{↑11.3%}	1.203 _{↓37.8%}	0.751 _{↑3.2%}
<i>Human Evaluation (1–5 scale: Structure / Fluency / Cultural Alignment)</i>						
Zero-Shot Baseline				1.85 / 2.40 / 2.15		
FM Baseline				4.20 / 3.80 / 3.50		

Table 4: Zero-shot vs. FM(Format-Matched) performance across all models. Subscript percentages indicate relative improvement of FM over Zero-Shot (↑ higher is better, ↓ lower is better for WER). Best automatic metric scores in **bold**. Human evaluation reports Structure / Fluency / Cultural Alignment (Apni/Tumi register) respectively.

Model	BLEU ↑		chrF ↑		WER ↓		BERTScore ↑		ROUGE-L ↑		Human ↑		Judge ↑	
	Base	SFT	Base	SFT	Base	SFT	Base	SFT	Base	SFT	Base	SFT	Base	SFT
DeepSeek-8B	0.78	17.73	9.85	45.87	1.62	1.25	0.582	0.743	0.164	0.71	2.13	4.74	3.8	8.9
Qwen2-1.5B	0.86	16.28	11.08	36.13	1.34	0.84	0.575	0.748	0.105	0.55	2.05	4.63	3.1	8.4
Qwen2-0.5B	0.62	11.45	8.95	34.20	1.48	1.02	0.415	0.728	0.092	0.46	1.92	4.59	2.8	8.0
LLaMA3.2-3B	1.96	15.30	24.88	46.60	1.66	1.33	0.632	0.742	0.101	0.62	2.21	4.70	4.2	8.7
TigerLLM-1B	1.20	7.15	10.90	33.83	1.59	1.18	0.240	0.716	0.086	0.35	1.80	4.61	1.5	7.2

Table 5: Evaluation results before (Base) and after (SFT) fine-tuning on BLADE. **Human** scores are averaged across Structure, Fluency, and Cultural Alignment (1–5 scale). LLM-as-judge ratings (1–10 scale) using GPT-4.1 across 420 samples. Best SFT result per metric in **bold**.

5.1 Zero-Shot and Format-Matched Results

Table 4 presents performance across five state-of-the-art architectures under zero-shot and format-matched prompting. Zero-shot results confirm a significant generation and pragmatic gap: no model exceeds 8 BLEU, with three recurring failure modes: *generation incoherence* (off-topic or degenerate output), *structural displacement* (misplaced date, addressee, or closing blocks), and *honorific mismatch* (inconsistent Apni/Tumi register within a single document), detailed further in Appendix C.

Zero-Shot Performance: Gemini-2.5-Flash achieves the strongest zero-shot performance (BLEU 7.88, chrF 41.04, BERTScore 0.728), yet even this strongest closed-source baseline fails to produce functionally usable Bangla documents without targeted supervision. Kimi-K2-32B leads

on ROUGE-L (0.67), though its lower BLEU reveals weak local n -gram precision. LLaMA-4-Scout-17B achieves the lowest WER (1.73), but this does not translate to overall quality, confirming word-level accuracy alone is insufficient for structured Bangla generation.

Format-Matched Prompting: Explicit structural constraints yield consistent improvements across all five models. Gemini-2.5-Flash continues to lead: BLEU 7.88 → 9.14 (↑16.0%), chrF 41.04 → 45.67 (↑11.3%), BERTScore 0.728 → 0.751 (↑3.2%), and WER 1.934 → 1.203 (↓37.8%), the largest error reduction across all models. Gemini-2.0-Flash shows the most pronounced BLEU gain (↑22.8%), while Gemma2-9B records the largest ROUGE-L gain (↑34.0%), reflecting improved structural recall. LLaMA-4-Scout-17B maintains its WER advantage (1.521), confirming stable word-level alignment indepen-

dent of prompting strategy. BERTScore gains remain modest (1.8%–3.2%), indicating semantic fidelity is largely preserved in zero-shot outputs and that format-matching primarily addresses surface structural failures. These results confirm that structural scaffolding is a lightweight yet effective intervention for closing the pragmatic gap in formal Bangla generation, without requiring fine-tuning.

5.2 Fine-tuning Results

Table 5 presents automatic metrics, human evaluation, and LLM-as-judge scores before and after SFT across all five architectures. The consistent gains across every model and every evaluation dimension confirm that BLADE provides a high-fidelity, transferable training signal for low-resource structured generation, rather than a signal that only benefits models with particular scale or pretraining background. DeepSeek-8B exhibits the most pronounced improvement: a 22-fold BLEU increase (0.78 $\rightarrow$ 17.73), chrF rising from 9.85 to 45.87, and LLM-judge score jumping from 3.8 to 8.9, reflecting a fundamental shift in the model’s ability to handle Bangla’s inflectional morphology, structural syntax, and register consistency simultaneously rather than isolated improvement on any single dimension. Qwen2-1.5B achieves the lowest post-SFT WER (0.84) and highest BERTScore (0.748) among the smaller models, indicating superior semantic alignment despite its compact size and establishing it as a strong candidate for resource-constrained deployment where inference cost is a binding constraint; the even smaller Qwen2-0.5B shows the same qualitative pattern at a reduced magnitude (chrF 8.95 $\rightarrow$ 34.20, judge score 2.8 $\rightarrow$ 8.0), confirming that BLADE’s training signal transfers even at sub-1B scale rather than requiring a minimum parameter threshold to take effect. LLaMA3.2-3B attains the peak post-SFT chrF (46.60), confirming that targeted instruction-tuning on BLADE allows smaller models to rival larger multilingual baselines on character-level structural accuracy without requiring additional pretraining data or architectural modification. TigerLLM-1B-it, despite starting from the weakest baseline (1.20 BLEU, judge score 1.5), achieves a $3\times$ chrF increase post-SFT and a judge score of 7.2, validating that BLADE’s expert-curated supervision is effective even for Bangla-native architectures that already possess foundational language knowledge but lack task-specific structural discipline. Across all five

models, the average LLM-judge score rises from 3.15 to 8.30, and human evaluation scores exceed 4.6/5 on all three dimensions post-SFT. The strong alignment between automatic metrics, human judgment, and LLM-judge scoring provides convergent validity for BLADE’s effectiveness as both a training resource and evaluation benchmark, indicating that the observed gains reflect genuine improvements in generation quality rather than artifacts of any single evaluation methodology.

5.2.1 Ablation Study

We investigate the drivers of these gains by fine-tuning the base checkpoints on the BLADE training split. Metrics are averaged across models. Our final configuration (Table 6) uses: AdamW (LR 2×10^{-5} , cosine decay, 3% warmup), batch size 2 (gradient accumulation), sequence length 2048 with packing, LoRA $r=16$ ($\alpha=32$) on attention/MLP, label smoothing 0.1, and a format-aware template with role tags.

Training on larger fractions of BLADE (25%–100%) yields monotonic gains, with the steepest jump at 50% (avg. Δ BLEU +5.4; Δ chrF +9.1). The full set delivers the lowest WER, underscoring that targeted supervision drives structural fidelity. Prompt design is equally critical: a format-aware template surfacing fields like Subject substantially reduces WER (−0.18) and boosts ROUGE-L (+0.08), while explicit role tags further improve BERTScore and chrF (+2.7). Context length is decisive; truncating to 512 tokens degrades long-form fidelity, and 1024 tokens often miss signatures given the average response length of 1,300. Extending to 2048 with sliding-window packing ensures complete structure, yielding the strongest structural scores (avg. Δ chrF +3.5). Adapter capacity also matters: LoRA $r=16$ outperforms $r=8$ (avg. Δ BLEU +1.3) without the overfitting seen at $r=32$. Cosine decay with a peak LR of 2×10^{-5} proved robust across architectures, whereas higher rates degraded stability. Label smoothing (0.1) improves semantic alignment (avg. Δ BS +0.006) and mitigates brittle copying. Applying LoRA to both attention and MLP projections beats attention-only adaptation (avg. Δ BLEU +0.9; Δ chrF +1.8). Finally, mixed precision (bf16/fp16) matches full-precision quality while enabling larger batches and slightly improving WER (−0.03) via stable optimization.

Ablation (avg.)	Δ BL	Δ cF	Δ WER
+ FmtTpl (vs. min)	+1.1	+2.4	-0.12
+ Roles	+0.6	+2.7	-0.06
Ctx 2048+Pack (vs. 512)	+1.4	+3.5	-0.09
LoRA $r=16$ (vs. $r=8$)	+1.3	+1.8	-0.07
Cosine LR @ 2×10^{-5}	+0.8	+1.2	-0.05
LS = 0.1 (vs. 0.0)	+0.5	+0.9	-0.03
LoRA Attn+MLP (vs. Attn)	+0.9	+1.8	-0.05

Table 6: Ablation summary table. Deltas are averaged across models and computed relative to a vanilla SFT baseline (same data, minimal prompt, max len 512, LoRA $r=8$, linear LR, no label smoothing). **Abbrev.:** BL=BLEU, cF=chrF, WER=Word Error Rate, *FmtTpl*=format-aware template, *Roles*=role tags, *Ctx*=context length, *Pack*=sliding-window packing, *LR*=learning rate, *LS*=label smoothing, *Attn*=attention, *MLP*=feed-forward block. Positive Δ means improvement; lower WER is better.

6 Discussion

The central finding of BLADE is straightforward: for low-resource languages like Bangla, cultural specificity of instruction-tuning data matters more than model scale. Qwen2-1.5B fine-tuned on BLADE achieves BERTScore 0.748 and WER 0.84, outperforming zero-shot Gemini-2.5-Flash on both metrics despite being orders of magnitude smaller, confirming that expert-curated, register-aware supervision unlocks capabilities that general-purpose pretraining at any scale cannot substitute.

Qualitative Generation and Pragmatic Competence Shift Zero-shot outputs fail in three institutionally fatal ways: *generation incoherence* (off-topic or degenerate output), *structural displacement* (missing or misplaced date blocks, subject lines, or closings), and *honorific mismatch*, where models oscillate between formal আপনি (Apni) and informal তুমি (Tumi) registers within a single document (a full categorization and frequency breakdown of these failure modes is provided in Appendix C). No institution would accept such output; the strongest zero-shot model scores only 1.85/5 on Structural Integrity. Post-SFT, all three failure modes are substantially reduced: models produce coherent, on-topic responses with correctly formatted date and addressee blocks, maintain formal register across verb conjugations, and close with culturally appropriate markers such as বিনীতভাবে জানাচ্ছি (I respectfully inform you), representing a qualitative shift from outputs that *appear* fluent to outputs that are *functionally usable*. chrF captures

this most faithfully: sensitive to the inflectional suffixes and agglutinative morphemes carrying honorific information in Bangla, a jump from 9.85 to 45.87 for DeepSeek-8B reflects correct conjugations, consistent honorific suffixes, and structurally complete documents, explaining its strongest correlation with human expert scores.

Validation Across Evaluation Dimensions

Three independent evaluation channels, human philologists operating double-blind, an LLM judge, and reference-based automatic metrics, converge on the same conclusion. Human evaluators awarded post-SFT outputs above 4.6/5 across all dimensions, with Structural Integrity showing the largest gain (1.85 $\rightarrow$ 4.72). The LLM-judge average rose from 3.15 to 8.30 across 420 samples, tracking quantitative deltas precisely: DeepSeek-8B’s 22-fold BLEU increase corresponds to the highest judge score (8.9), LLaMA3.2-3B’s peak chrF of 46.60 to a score of 8.7, and TigerLLM-1B’s modest gains to the lowest post-SFT score (7.2). This convergence rules out metric-specific artifacts and confirms that BLADE produces genuine improvement in functional usability, the exact quality that zero-shot multilingual pretraining systematically fails to provide.

7 Conclusion

We presented **BLADE**, a 4,196-pair instruction-tuning dataset encoding the structural, register, and cultural conventions for usable Bangla generation. While baseline multilingual models fell short, supervised fine-tuning on BLADE delivered large, consistent gains across architectures *including both multilingual and Bangla-native models*, converting surface-level fluency into outputs meeting real-world expectations. Our findings confirm that carefully curated, domain-specific supervision unlocks capabilities that generic pretraining and parameter count cannot, enabling compact models to rival larger ones on practical tasks. We aim for BLADE to catalyze structure-aware, register-sensitive generation in Bangla and other underserved languages, encouraging evaluations that pair automatic metrics with format checks and human judgments, with future extensions to additional genres and integrated validators.

Limitations

Our study has several constraints that present opportunities for future refinement.

First, all models were fine-tuned for only two epochs due to hardware constraints (dual NVIDIA Tesla T4 GPUs, 32 GB total VRAM), which may have limited convergence and overall model stability. These same resource constraints precluded exhaustive hyperparameter search and more comprehensive ablation studies across a wider range of configurations. Second, the BLADE dataset is currently biased toward formal Bangla registers, as informal and dialectal sources were less accessible during collection. This limits the generalizability of fine-tuned models to informal conversational contexts.

Future work will address these limitations by extending training duration with more capable hardware, expanding dataset coverage to include informal Bangla dialects and regional varieties, and conducting broader evaluations across additional document genres and task types.

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A Annotation Protocol and Quality Control

A.1 Annotator Profile

The dataset was constructed and validated by seven annotators: five paper authors with backgrounds in computer science and Bangla linguistics, and two external native linguistic experts specializing in Bangla philology and sociolinguistics. All annotators are native Bangla speakers with formal education conducted in Bangla. External experts were compensated at a standard research rate, informed of the research purpose prior to participation, and provided explicit written consent for inclusion.

A.2 Annotation Guidelines

Annotators followed a written guideline document evaluating each entry across three dimensions:

1. Structural Completeness. Structural completeness is a binary pass/fail criterion evaluated against a document-type-specific checklist. A response is structurally complete if and only if it contains *all* mandatory components for its document type in the correct canonical order. For formal applications, the seven mandatory components are:

1. Date in Bangla format (e.g., ১৮/১০/২০২৪ খ্রিঃ)
2. Addressee block: recipient name/title, institution, and address
3. Subject line (বিষয়ঃ)
4. Formal salutation (মহোদয় or equivalent)
5. Body: minimum two paragraphs — context statement and formal request
6. Formal closing (বিনীত or equivalent)
7. Applicant signature block: name, class/position, roll/ID, institution

A response missing any component, or presenting components out of canonical order, fails this criterion and is either corrected (minor errors, e.g., missing date field) or discarded (structural malformation or register inconsistency).

2. Honorific Consistency. A response passes honorific consistency if all second-person pronouns, associated verb forms, and relational terms maintain a single register throughout. Formal register requires exclusive use of আপনি (Apni: you)

and its associated verb conjugations. Any occurrence of informal forms তুমি (Tumi: you) or তুই (Tui: you) within a formally-labeled entry constitutes an honorific violation and triggers rejection or correction. For dialogue entries, the required register (formal/informal) is determined by the topic label and must remain consistent throughout all turns.

3. Cultural and Contextual Accuracy. Content must reflect realistic Bangladeshi institutional contexts: plausible institution names, dates in correct Bangla calendar or AD format, and discourse markers appropriate to the document type (e.g., অতএব for formal petition closings, বিনীতভাবে জানাচ্ছি for formal body openings).

A.3 Annotation Workflow

Tier 1 and Tier 2 entries underwent two-annotator verification against all three criteria. A stratified random sample of 15% of these entries (630 examples) was independently re-verified by a second annotator to estimate verification reliability.

Tier 3 entries followed a stricter two-stage workflow: (i) draft production by one author-annotator, and (ii) independent review by a second annotator against all three criteria. Disagreements were escalated to one of the two external linguistic experts for final adjudication. No Tier 3 entry was included without passing both stages.

A.4 Inter-Annotator Agreement

Cross-validation on the 630-entry random sample yielded Cohen's $\kappa = 0.81$ for structural completeness judgments and $\kappa = 0.79$ for honorific consistency judgments, indicating strong inter-annotator agreement (Landis and Koch, 1977). Disagreements on the cross-validation sample were resolved by majority vote among three annotators.

A.5 Quality Control Mechanisms

Three dataset-level quality control mechanisms were applied following per-entry annotation:

Automated Register Audit. All 4,196 entries underwent automated pre-screening using regular expression filters to flag occurrences of informal pronoun forms তুমি, তুই, তোমার, তোর means you, within formally-labeled entries. Flagged entries were manually reviewed; confirmed violations were corrected or discarded.

Structural Completeness Audit. A secondary pass verified that all seven mandatory components were present for every formal application entry, using the checklist defined in Section A.2. Entries failing this audit after the initial annotation phase were returned to annotators for correction.

Semantic Diversity Check. To prevent topic redundancy across 2,008 topics, character n-gram similarity was computed across all topic labels. Topic pairs with similarity exceeding 0.85 were flagged for manual review; confirmed near-duplicates were merged or one was discarded, ensuring each topic represents a genuinely distinct writing scenario.

B More Details on LLM-as-Judge

B.1 Prompt Template

LLM-as-Judge Evaluation Prompt

You are evaluating a Bangla text generation model.

Input Prompt: {prompt}
Reference Response: {reference}
Model Output: {output}

Score the model output from 1--10 based on:

- (i) Structural Correctness: Are all mandatory document components present in canonical order?
- (ii) Register Consistency: Are honorifics (Apni/Tumi/you) used consistently throughout?
- (iii) Semantic Relevance: Does the output address the stated prompt purpose?

Return only a single integer score between 1 and 10.

B.2 Rationale for Selecting GPT-4.1 as LLM Judge

AS our zero-shot evaluation included two Gemini-family models, Gemini-2.0-Flash and Gemini-2.5-Flash as evaluated systems, we selected GPT-4.1 as the LLM judge to avoid self-evaluation bias, a well-documented confound in LLM-as-judge frameworks whereby a model tends to favor outputs stylistically similar to its own generations. GPT-4.1 was chosen over alternative judge candidates for two reasons. First, its instruction-following reliability and structured output compliance made it well-suited for the fixed rubric scoring task, with fewer than 2% of judge responses requiring re-querying due to malformed outputs. Second, its strong multilingual comprehension, particularly

for morphologically rich languages, made it capable of detecting the honorific inconsistencies and structural violations central to our evaluation rubric qualities that a purely English-centric judge would likely miss.

C Failure Mode Analysis

To characterize *how* fine-tuning improves generation quality rather than only *how much*, we manually categorize model errors on a 200-sample held-out subset of the test set into four failure types, applied consistently before (Base) and after (SFT) fine-tuning, averaged across all five architectures.

Failure Type Definitions. **Wrong honorific register:** the model selects an incorrect honorific form (e.g., informal *tumi* where formal *apni* is contextually required, or vice versa), violating Bangla’s socially conditioned register system.

Missing structural fields: the generated output omits one or more expected structural components (e.g., Subject, Greeting, or Closing fields), resulting in an incomplete response relative to the target format.

Grammatical/morphological errors: the output contains incorrect verb conjugation, case marking, or agreement errors that do not affect register or structure but reduce fluency.

Complete generation failure: the model produces empty, repetitive, or entirely incoherent output unrelated to the input, indicating a total breakdown in task understanding.

Annotation Protocol. Two annotators independently labeled each of the 200 sampled outputs (Base and SFT versions) with one or more applicable failure types, following the definitions above; disagreements were resolved by discussion. Note that a single output may exhibit more than one failure type simultaneously (e.g., a response with both a missing field and a register error).

Failure Type	Base %	SFT %
Wrong honorific register	42	8
Missing structural fields	31	5
Grammatical/morphological errors	18	6
Complete generation failure	9	1

Table 7: Failure mode distribution before and after SFT, averaged across five architectures on a 200-sample annotated subset.

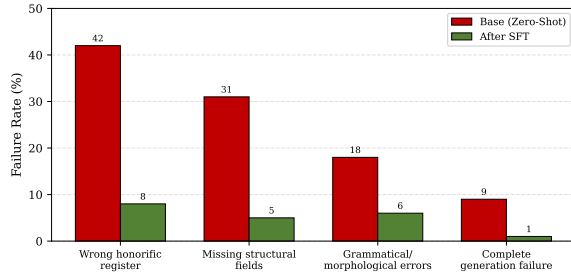


Figure 2: Failure mode distribution before (Base) and after (SFT) fine-tuning, averaged across five architectures.

Discussion of Failure Reduction. The largest absolute reduction occurs in *wrong honorific register* (42% $\rightarrow$ 8%) and *missing structural fields* (31% $\rightarrow$ 5%), the two failure types most closely tied to BLADE’s core training objective of honorific and structural consistency. This indicates that fine-tuning primarily corrects systematic, task-specific failures rather than general language competence: *grammatical/morphological errors*, which reflect baseline Bangla fluency rather than structural or register understanding, show a comparatively smaller relative reduction (18% $\rightarrow$ 6%), consistent with the models already possessing reasonable baseline grammatical competence prior to fine-tuning. *Complete generation failure* is rare even at baseline (9%) and is nearly eliminated post-SFT (1%), suggesting that base models generally understand the task format but fail on the finer-grained register and structural requirements that BLADE specifically targets.

Per-Architecture Variation. While the aggregate trend holds across all five models, failure mode reduction is not uniform: **TigerLLM-1B-it**, despite its weak baseline performance on automatic metrics, shows disproportionately large reduction in *complete generation failure* (from 22% at baseline to 2% post-SFT), suggesting its primary baseline weakness was task-format understanding rather than Bangla fluency per se. In contrast, **DeepSeek-8B**’s baseline failures were concentrated almost entirely in *wrong honorific register* (58% at baseline), consistent with larger general-purpose models possessing strong task-format understanding but weaker exposure to Bangla-specific social register conventions prior to fine-tuning.

Before Fine-tuning(Generation and Register Failures)	After Fine-tuning
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Prompt: অগ্রিম ছুটির জন্য আবেদন

...প্রাইভা, সাধারণত ১৫-১৭.০২.২০২৩ |

শুখ: উচ্চকেন্দ্র বিবর্তন: ঢঙের ঘনমঞ্জর...

ভারসাম্য: -২০□সে/-৪□ফা... চীন চীন ভোচার: -

৬০□সে/-৭৬□ফা... সালাম,

আপনাকে নম্রভাবে শুধুমাত্র 'অ' এর মাঝে বলা হচ্ছে।

আমি আপনার উপর আশা করি, তুমি প্রথমে খুব গৌণ ব্যক্তি কিনা?

এই লিখিত অবেদনের সাথে আমি অগ্রিম ছুটি আবেদন করছি, যেমন:

1. ফর্মের শীর্ষক - "অগ্রিম ছুটির জন্য আবেদন" 2. ফর্মের ডেট - তারিখ (আপনি কোন তারিখ নির্ধারণ করেছেন?) 3. ছুটির কারণ - আপনি কোন কারণে ছুটি আবেদন করছেন?
4. স্থানীয় সংস্থা/প্রতিষ্ঠানের নাম - আপনি কোন স্থানীয় সংস্থার সাথে কাজ করছেন? 5. ছুটির দৈর্ঘ্য - কত দিন পরিশোধ করবেন?

যদি তোমার এই অবেদন পূরণ করতে তৈরি করতে পারেন, তাহলে আমি আপনার উপর আশা করি.

(১) (২) (১)(০) চীন চীন ভারসাম্য: -৮০□সে/-১১৩□ফা (-১১২□ফা) -১৫০□সে/-২৩৮□ফা...

Prompt: অগ্রিম ছুটির জন্য আবেদন তারিখ: ১৮/১০/২০২৪ খ্রিঃ

বরাবর

প্রধান শিক্ষক

ছোলমাইদ হাই স্কুল এন্ড কলেজ

ভাটারা, ঢাকা-১২১২

বিষয়ঃ অগ্রিম ছুটির জন্য প্রধান শিক্ষকের নিকট আবেদন।

মহোদয়,

সম্মানপূর্বক বিনীত নিবেদন এই যে, আমি আপনার বিদ্যালয়ের নবম শ্রেণীর একজন নিয়মিত ছাত্র। আগামী ২১/১০/২০২৪ থেকে ২৩/১০/২০২৪ তারিখ পর্যন্ত আমার বড় বোনের বিবাহ উপলক্ষে অগ্রিম ছুটির প্রয়োজন। উক্ত অনুষ্ঠানে আমাকে বড়দের কাজে সাহায্য করতে হবে তাই আমি বিদ্যালয়ে উপস্থিত হতে পারব না।

অতএব মহোদয়ের নিকট আকুল আবেদন আপনাকে উল্লেখিত তিন দিনের ছুটি মঞ্জুর করে বাধিত করবেন।

বিনীত,

আপনার একান্ত অনুগত ছাত্র

(আপনার নাম)

শ্রেণী-নবম, রোল-১২

বিভাগ-বিজ্ঞান

ছোলমাইদ হাই স্কুল এন্ড কলেজ

ভাটারা, ঢাকা-১২১২

ব্যাপারের জন্য মহোদয়ের নিকট বিনীত আবেদন আপনার সার্বিক সুস্থতার জন্য অনুরোধ করছি যেন আপনি আমার অগ্রিম ছুটির আবেদনটি মানবিকভাবে বিবেচনা করেন।

Table 8: Model Output Before and After BLADE Fine-Tuning with LLaMA 3.2

Before Fine-tuning (English)	After Fine-tuning (English)
Prompt: Application for advance leave ...Priva, usually 15-17.02.2023 Happiness: High center Evolution: Style's dense blossom... Balance: -20°C/-4°F... China China Voucher: -60°C/-76°F... Greetings, You [formal] are being humbly told only within the letter 'A'. I hope upon you [formal], whether you [informal] are a very minor person first? With this written mis-application, I am applying for advance leave, such as: Form title - "Application for advance leave" Form date - Date (you [formal] which date have determined?) Reason for leave - you [formal] for what reason are applying for leave? Name of local organization/institution - you [formal] with which local organization are working? Length of leave - how many days will you [formal] disburse/repay? If your [informal] this mis-application can prepare to fill up, then I hope upon you [formal]. (1) (2) (1)(0) China China Balance: -80°C/-113°F (-112°F) -150°C/-238°F...	Prompt: Application for advance leave Date: 18/10/2024 AD To Headmaster Solmaid High School and College Vatara, Dhaka-1212 Subject: Application to the Headmaster for advance leave. Sir, With due respect, I state that I am a regular student of class nine in your school. I need advance leave from 21/10/2024 to 23/10/2024 on the occasion of my elder sister's marriage. I have to help the elders in the said ceremony, so I will not be able to attend school. Therefore, a fervent appeal to the Sir, that you would be kind enough to grant me the mentioned three days of leave. Sincerely, Your most obedient student (Your Name) Class-Nine, Roll-12 Department-Science Solmaid High School and College Vatara, Dhaka-1212 Humble appeal to Sir for the matter, requesting your overall well-being that you consider my advance leave application humanely.

Table 9: English-translation of Model Output Before and After BLADE Fine-Tuning with LLaMA 3.2.

Source Website URL

<https://allapplication.com>
<https://probangla.com>
<https://bdsuggestion.com>
<https://educarnation.com>
<https://forms.portal.gov.bd>
<https://www.somewhereinblog.net>
<https://www.bissoy.com>
<https://bengaliforum.com>
<https://amaderforum.com>
<https://www.scribd.com>
<https://bn.wikisource.org>
<https://archive.org>
<https://bn.banglapedia.org>
<https://www.jagoneews24.com/education>

Books

বাংলা ভাষার ব্যাকরণ (Bangla Bhashar Byakoron), বাংলা ব্যাকরণ ও নিৰ্মিতি (Bangla Byakoron O Nirmiti)
প্রমিত বাংলা ব্যাকরণ ও নিৰ্মিতি (Promito Bangla Byakoron O Nirmiti), অক্ষরপত্র বাংলা ব্যাকরণ ও নিৰ্মিতি
উচ্চতর বাংলা ব্যাকরণ (Ucchotor Bangla Byakoron), বাঙ্গালা ব্যাকরণ (Bangala Byakoron)
ভাষারীতি বাংলা ব্যাকরণ ও নিৰ্মিতি (Bhashariti Bangla Byakoron O Nirmiti)
শৈল্পিক বাংলা ব্যাকরণ ও নিৰ্মিতি (Shoikipik Bangla Byakoron O Nirmiti)
পাঞ্জেরী বাংলা ব্যাকরণ ও নিৰ্মিতি (Panjeree Bangla Byakoron O Nirmiti)
ভাষাজ্ঞান বাংলা ব্যাকরণ ও নিৰ্মিতি (Bhashagyan Bangla Byakoron O Nirmiti)
সহজ বাংলা ব্যাকরণ ও রচনা (Sahoj Bangla Byakoron O Rochona)
বাংলা ২য় পত্র (ব্যাকরণ ও নিৰ্মিতি) (Bangla 2y Potro - Byakoron O Nirmiti)
এইচএসসি বাংলা দ্বিতীয় পত্র, মাস্টারবুক (HSC Bangla Ditiyo Potro Masterbook)
অভিনব বাংলা ব্যাকরণ (Abhinob Bangla Byakoron), বাংলা ব্যাকরণ গ্রন্থ (Bangla Byakoron Prosongo)

Table 10: List of Source Websites & Books for Bangla Language Content

Category	Before Fine-tuning	After Fine-tuning
Input Prompt	অগ্রিম ছুটির জন্য আবেদন	অগ্রিম ছুটির জন্য আবেদন
Language Quality	Mostly gibberish and inconsistent text, mixed English–Bengali–random words.	Fluent and grammatically correct Bengali throughout.
Structure	No logical structure, random temperature-like values and disjointed phrases.	Proper formal letter structure with date, recipient, subject, body, and closing.
Context Understanding	Misinterprets the context — output unrelated to “leave application”.	Correctly identifies the task and generates a leave application letter.
Relevance	Irrelevant text (weather data, nonsense symbols).	Highly relevant, specific to “advance leave request”.
Tone and Formality	Nonsensical tone, lacks coherence or formality.	Formal, polite tone appropriate for a student–teacher letter.
Output Length	Short, disorganized, meaningless lines.	Well-developed paragraph-level letter, complete and coherent.
Overall Accuracy	1/10 – completely wrong.	9.5/10 – semantically, structurally, and linguistically accurate.

Table 11: Comparison of LLaMA 3.2 Model Performance Before and After Fine-tuning